

Welcome to Astronomy Lab projects

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Analysing gravitational wave data

Astronomy 345 Labs

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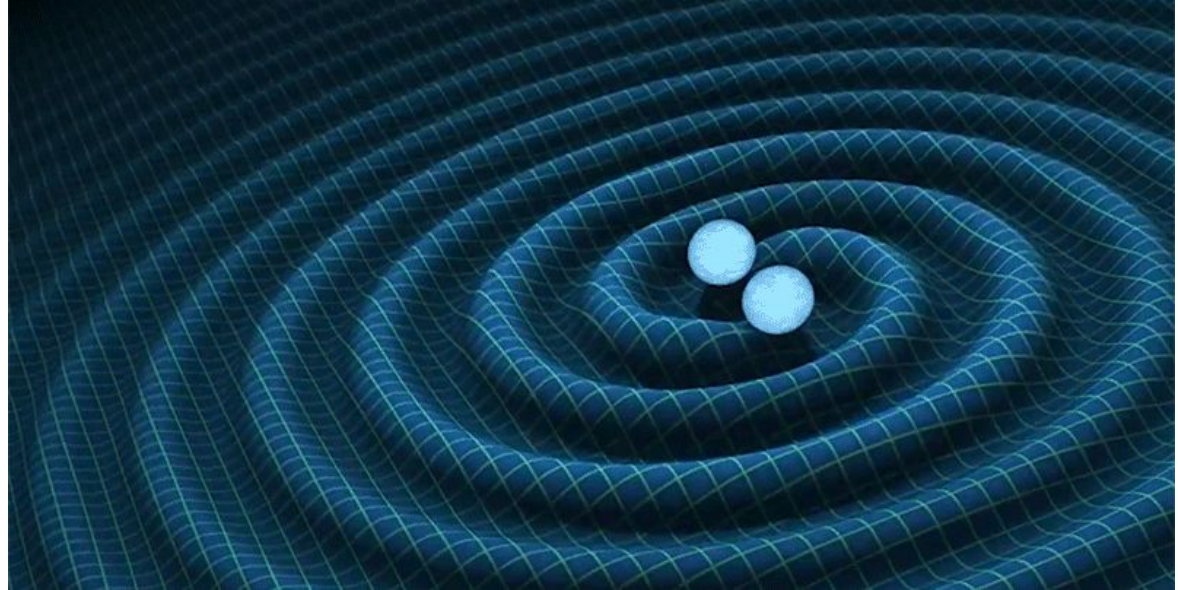
Ollie Burke

Introduction to gravitational waves

Different sources of gravitational waves

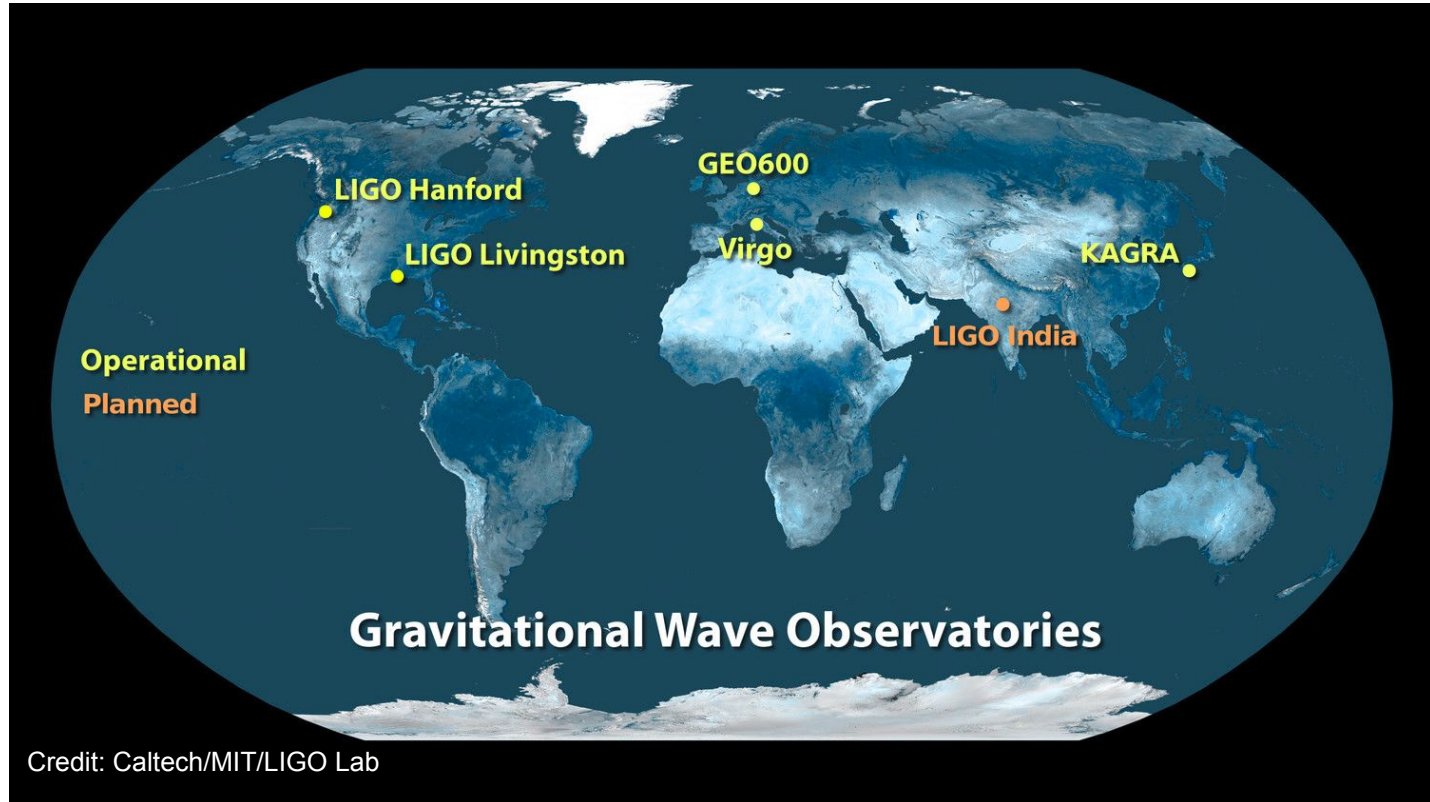
It takes very violent events to create gravitational waves, such as:

- Colliding black holes or neutron stars
- Exploding stars (supernovae)
- Rapidly rotating neutron stars with small mountains

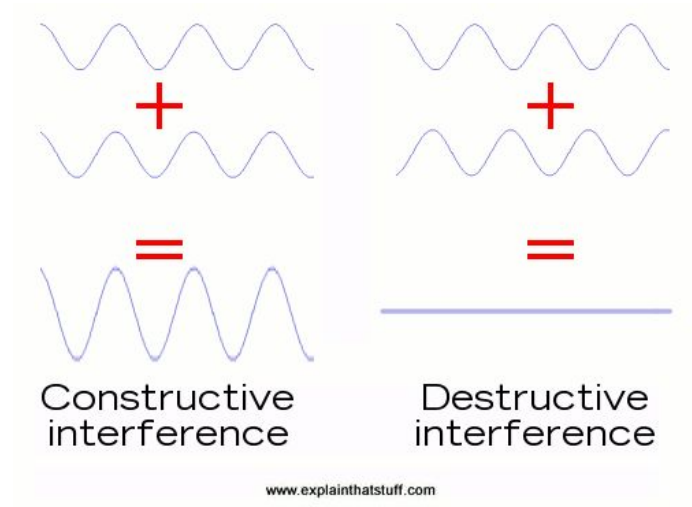
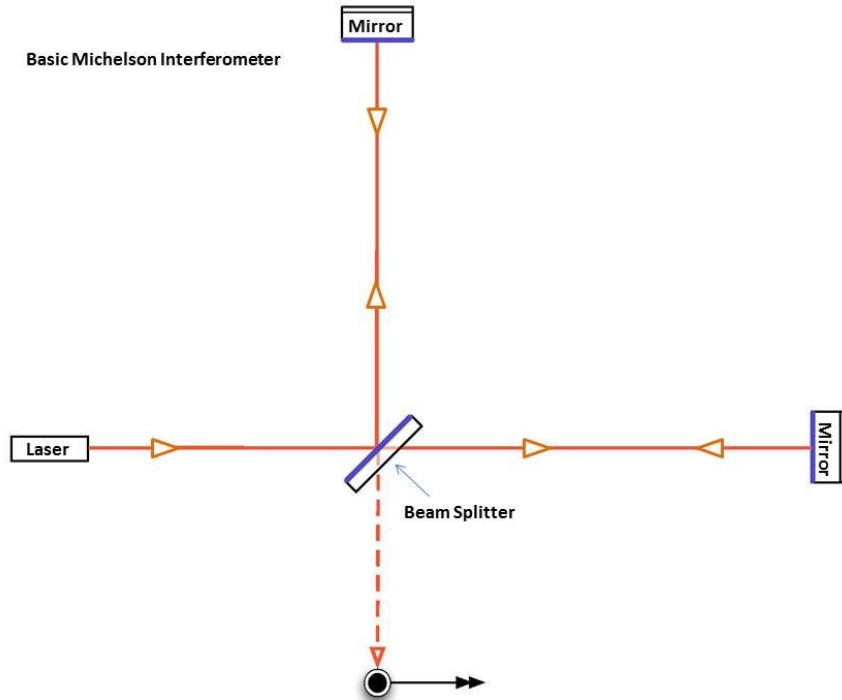


Credit: R. Hurt - Caltech / JPL: <https://www.ligo.caltech.edu/video/gravitational-waves>

Current gravitational wave observatories

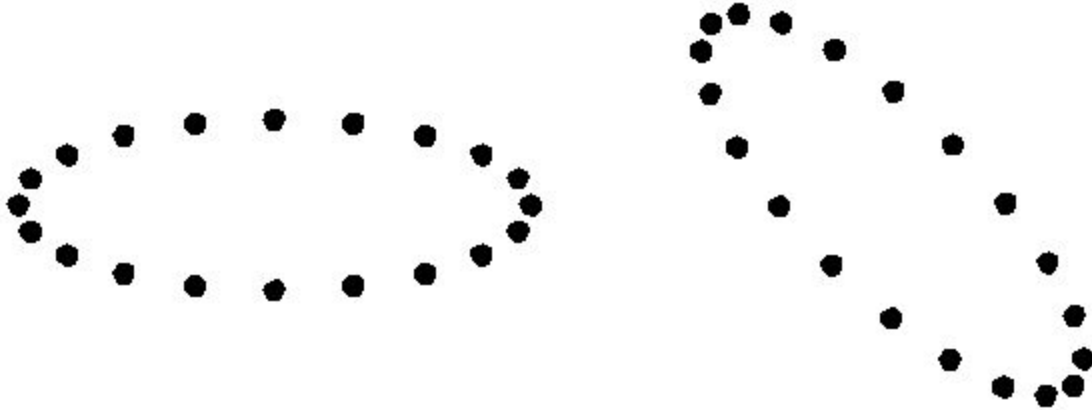


A gravitational wave interferometer



How a gravitational wave warps space-time

Plus (+) and cross (x) polarised gravitational waves



Directional sensitivity

Interferometers are not equally sensitive to GW signals in all directions.

Maximally sensitive perpendicular to the plane of the interferometer (directly overhead/below), and minimally sensitive in the same plane as the interferometer.

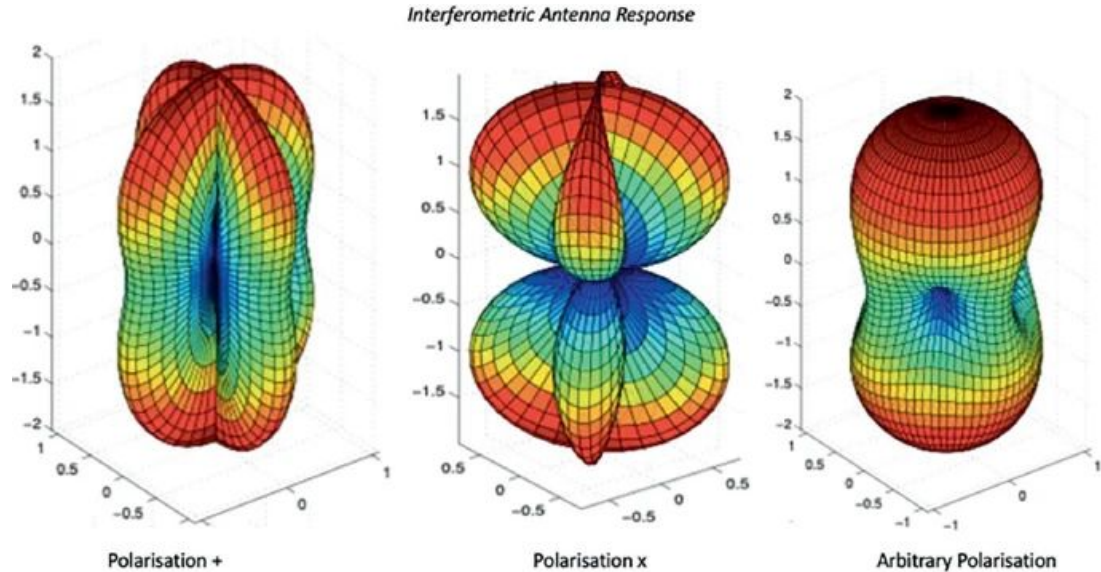


Image credit: Ricci, F., Bassan, M. (2022). Interferometric Detectors of Gravitational Waves.

Strain

The output of a gravitational wave detector, $d(t)$ is a combination of:

- Signal, $h(t)$ - the motion of the mirrors due to passage of a GW event.
- Noise, $n(t)$ (often assumed to be Gaussian)

$$d(t) = n(t) + h(t)$$

Where

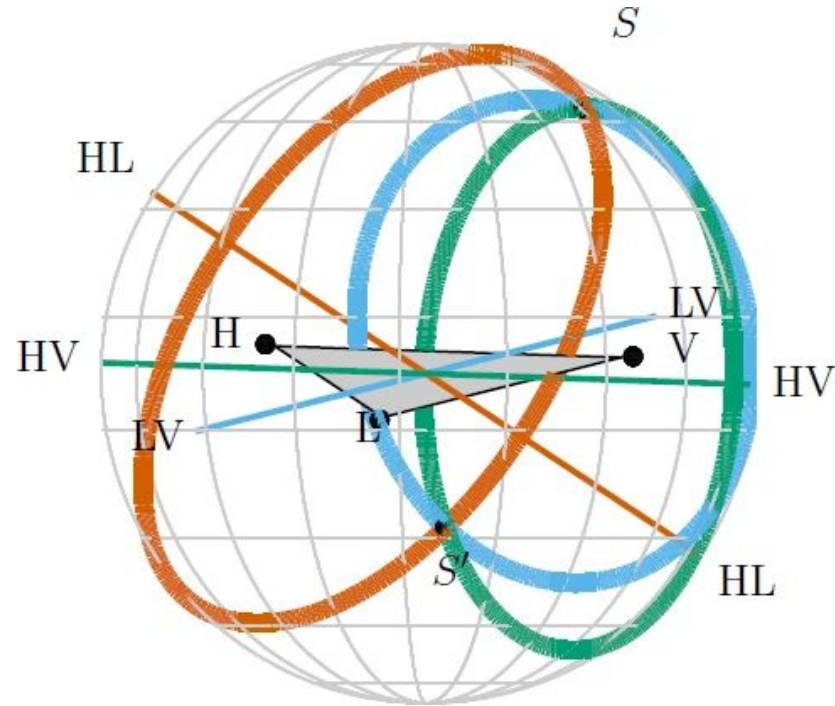
$$h(t) = F_+ h_+(t) + F_x h_x(t)$$

F_+ and F_x are the antenna response functions of the detector, dependent on the source sky position (relative to the detector) and polarisation of the waves

Benefits to multiple gravitational wave detectors

In reality the data from multiple detectors is analysed simultaneously for detection and parameter estimation:

- Better able to distinguish between real signals and glitches (real signals are common between detectors, while glitches are not).
- Higher SNR (can combine SNRs in quadrature).
- Improved localisation of sources on the sky due to triangulation.



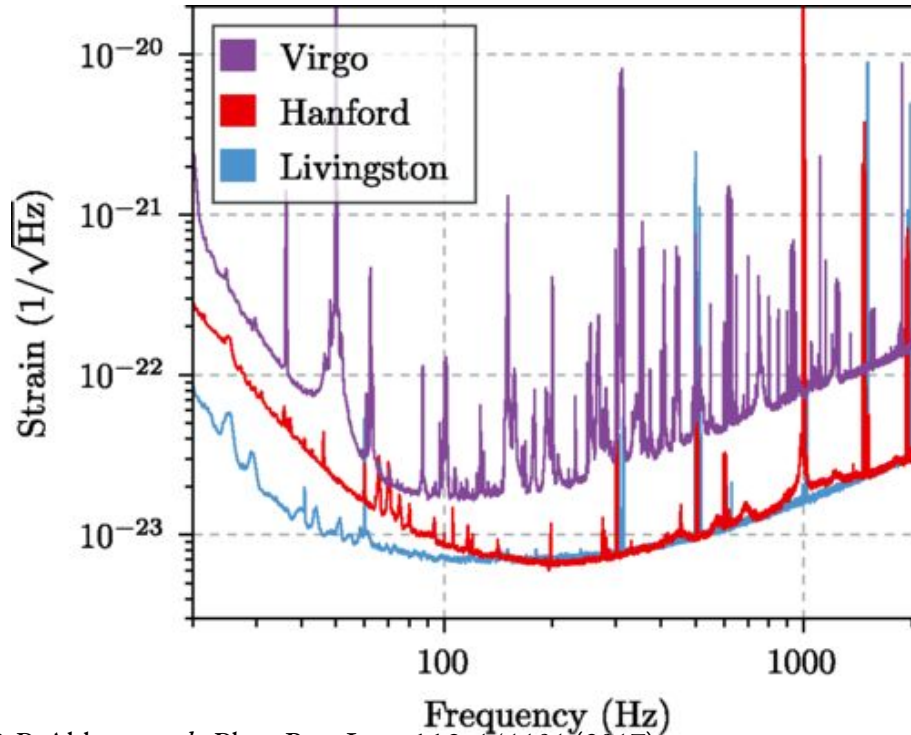
Time-domain vs frequency domain

Strain data is provided in the time domain (a list of the time of measurement and the measured strain at each time).

It is often useful to convert the signal into the frequency domain for efficiency, for which Fourier Transforms can be used.

Looking at the frequency domain can help us to better understand the properties of our noise and signal.

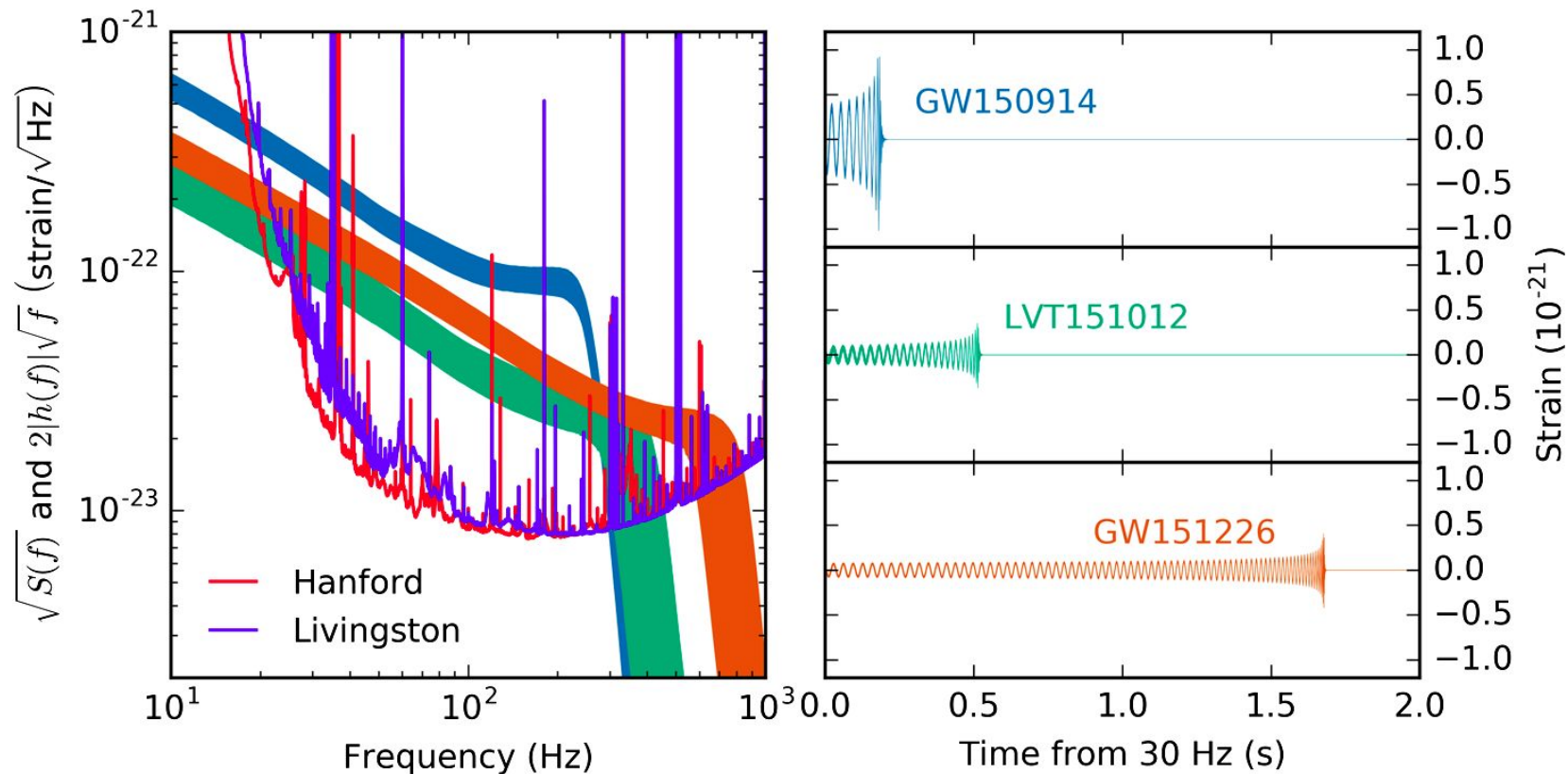
Sensitivity of a gravitational wave detector



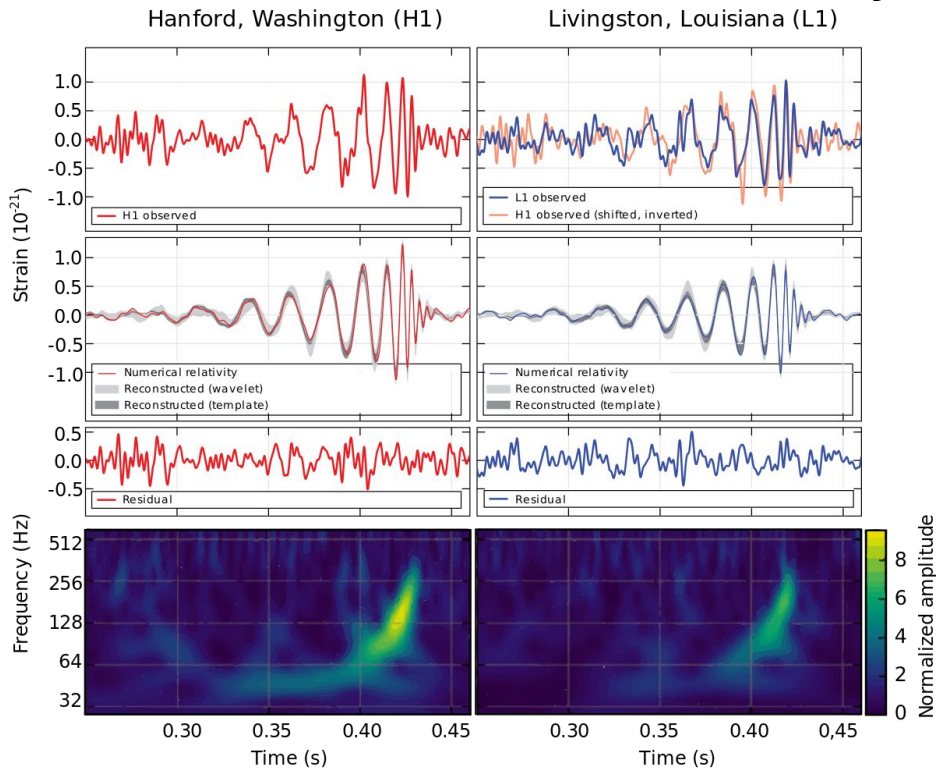
This plot shows the amplitude spectral density (ASD). This is related to the power spectral density (PSD) by: $\text{PSD} = \text{ASD}^2$

If you are working with “white noise” (as you will at the start of this lab), the power is equal across all frequency bins.

When working with real GW data, whitening is one of the first steps.



The first detection of a binary black hole



Two black holes, weighing 36 and 29 Msol, collided at a luminosity distance of 410 Mpc.

The signal they produced was detected on the 14th September 2015 by the two LIGO detectors.

B. P. Abbott et al. (LIGO Scientific Collaboration and Virgo Collaboration)

Phys. Rev. Lett. **116**, 061102 – Published 11 February 2016

Observing runs

Updated
2025-11-15

LIGO

Virgo

KAGRA

O1

80
Mpc

O2

100
Mpc

30
Mpc

O3

100-140
Mpc

40-60
Mpc

0.7
Mpc

O4

150-160+
Mpc

50-60
Mpc

1-3 Mpc ≈ 10 Mpc

O5

240-325
Mpc

70-130
Mpc

25-128
Mpc

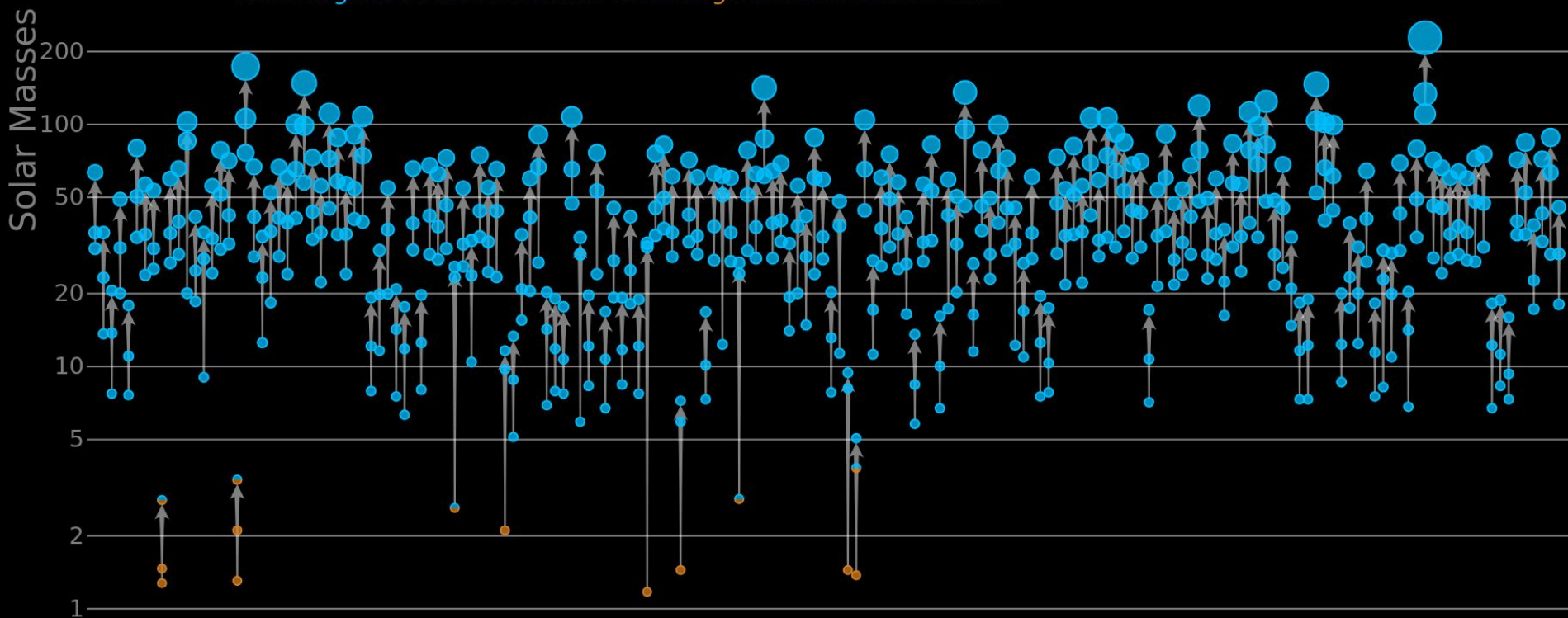
G2002127-v33

2016 2017 2018 2019 2020 2021 2022 2023 2024 2025 2026 2027 2028 2029 2030 2031

Credit: LIGO/Virgo/KAGRA

Masses in the Stellar Graveyard

LIGO-Virgo-KAGRA Black Holes *LIGO-Virgo-KAGRA Neutron Stars*



LIGO-Virgo-KAGRA | Aaron Geller | Northwestern

Lab structure

What's going to happen in this lab?

This lab is a “choose your own adventure” with two different paths:

Gravitational wave detection

How do we know when we've detected a gravitational wave? How do you know it's a real signal?

- Learn how to detect signals in a stream of data using matched filtering
- Detect the hidden signals in our mock data challenge

Gravitational wave parameter estimation

Where have these signals come from? What can we learn about them?

- Learn how to measure parameters such as masses and distance using Bayesian statistics and MCMC
- Find out more about the astrophysical source

What happens now?

There are 3 jupyter notebooks for this lab: [signal_processing.ipynb](#), [signal_detection.ipynb](#) and [bayesian_inference.ipynb](#).

Everyone will need the tools that are introduced in [signal_processing.ipynb](#), so look through this first.

Afterwards, choose whether you want to focus on *detection* or *parameter estimation* for the rest of the lab, and start working your way through the respective notebook

We encourage working in pairs and having 1 person focus on detection, and 1 person focus on parameter estimation during the lab. However this is not essential.

What happens now?

Depending on how quickly you get to grips with the material, you should reach part 4 of the mock data challenge when you still have several lab sessions remaining.

From this point on the lab is open ended!

- If you have worked on parameter estimation, you could try analysing the signals found by someone working on the detection part of the lab
- Or add another detector into your analysis to improve how well you can detect or infer parameters
- Or try out your code on some real gravitational wave data
- Or...

Timeline

- **Lab 1:** Read through material, choose your path (detection or parameter estimation) and start working through notebooks.
- **Lab 2:**
- **Lab 3:**
- **Lab 4:** By end of lab 4, you should be working on parts 2 or 3 in the notebooks and have some ideas about where to go next.
- **Interim reports due by the Fri 6th Feb**
- **Lab 5:** Chat with lab leads (Rachel and Joe)
- **Lab 6:**
- **Lab 7:**
- **Lab 8:**
- **Lab 9:** Final lab, make sure to have asked all your questions about analysis and report writing